

Tilings, Colouring and Enumeration

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Problems

Cornerless chessboard) Can an 8×8 board with two of its opposite corners removed be covered by dominoes?

Cornerless chessboard ver2) Can an 8×8 board with 1 corner removed be covered by 1×3 dominoes?

SerbianMO2023P2) A $2021 \times 2021 \times 2021$ cube is given. In how many ways can you stick on this cube a $1 \times 1 \times 1$ cube such that the resulting body can be tiled by $1 \times 1 \times k$ cuboids for some $k \geq 2$?

ISL2014C4) Construct a tetromino by attaching two 2×1 dominoes along their longer sides such that the midpoint of the longer side of one domino is a corner of the other domino. This construction yields two kinds of tetrominoes with opposite orientations. Let us call them S - and Z -tetrominoes, respectively. Assume that a lattice polygon P can be tiled with S -tetrominoes. Prove that no matter how we tile P using only S - and Z -tetrominoes, we always use an even number of Z -tetrominoes.

MGcup2023P7) We have an $m \times n$ board with some dominoes placed on it such that no two touch (having a common vertex is considered touching). Depending on m and n , is it always possible to tile the rest of the board with dominoes?

SerbianTST2020P3) On a $2n \times 2n$ board we place without overlapping k rectangles of dimensions $1 \times n$. If we know that no more such rectangles can be placed, find the minimal value of k .

SerbianMO2018P5) Let a, b be odd natural numbers greater than 1. Observe an $a \times b$ board with the fields $(2, 1)$, $(a - 2, b)$ and (a, b) missing. We tile such a board with 2×1 dominoes and 2×2 squares without overlapping. Prove that at least $\frac{3}{2}(a + b) - 6$ dominoes were used for the tiling.

USATST2009P1) Let m and n be positive integers. Mr. Fat has a set S containing every rectangular tile with integer side lengths and area of a power of 2. Mr. Fat also has a rectangle R with dimensions $2^m \times 2^n$ and a 1×1 square removed from one of the corners. Mr. Fat wants to choose $m + n$ rectangles from S , with respective areas $2^0, 2^1, \dots, 2^{m+n-1}$, and then tile R with the chosen rectangles. Prove that this can be done in at most $(m + n)!$ ways.

EGMO2022P5) For all positive integers n, k , let $f(n, 2k)$ be the number of ways an $n \times 2k$ board can be fully covered by nk dominoes of size 2×1 . (For

example, $f(2, 2) = 2$ and $f(3, 2) = 3$.) Find all positive integers n such that for every positive integer k , the number $f(n, 2k)$ is odd.

EGMO2019P2) Let n be a positive integer. Dominoes are placed on a $2n \times 2n$ board in such a way that every cell of the board is adjacent to exactly one cell covered by a domino. For each n , determine the largest number of dominoes that can be placed in this way. (A domino is a tile of size 2×1 or 1×2 . Dominoes are placed on the board in such a way that each domino covers exactly two cells of the board, and dominoes do not overlap. Two cells are said to be adjacent if they are different and share a common side.)